

UNIVERSITEIT VAN PRETORIA
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Faculty of Engineering, Built Environment & Information Technology

Department of Computer Science

COS326 - Database Systems

Online Semester Test

Total Marks: 100

16 September 2024

Time/Duration: 17:30pm - 19:00pm (1 Hour 30 Minutes)

Examiner: Mr. S.M. Makura

External Moderator: Mr F. Elegbeleye (WSU)

Initials and Surname: _____

Student Number: _____

Degree: _____

Instructions

1. Fill in the semester test paper with your details above.
2. Read the question paper carefully and answer all the questions.
3. This paper comprises of 26 questions on 29 pages. It consists of three sections,

namely Section A (Multiple Choice & Fill in the blank questions) - 20 marks, Section B (Application and Theory Based Questions) - 40 Marks, Section C (Scenario Based questions) - 40 Marks.

4. Answer **all** the questions in the answer book provided. After you are done, submit **both** the semester test paper together with your answer book.
5. You have 90 minutes to complete this semester test. Spend approximately the following minutes per each section:
   - Section A - 15 minutes
   - Section B - 30 minutes
   - Section C - 45 minutes
6. This is a closed book examination: you may therefore not have any study material with you.
7. Please switch off your smartphone/tablet and keep it off for the duration of the semester test.
8. All examination regulations and 'code of conduct' of the University of Pretoria are applicable during this semester test.

Integrity Statement:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Section A: Multiple Choice and Fill in the Blanks Questions (20 Marks)

(Spend approximately 15 minutes in this section)

Question 1

Which of the following is **NOT** one of the atomic query expressions in JPA (Java Persistence API)?

a. JPQL Variables

b. JPQL Parameters

c. JPQL Literals

d. JPQL Constants

Question 2

Given the following entity class definitions in a JPA application:

```
@Entity
public class Product {
    @Id
    private Long id;
    private String name;
    private double price;

    @ManyToOne
    private Supplier supplier;

    // Getters and setters
}

@Entity
public class Supplier {
    @Id
    private Long id;
    private String name;
    // Getters and setters
}
```

Which of the following JPA code snippets correctly retrieves a list of Product entities and their associated Supplier names, whose price is greater than 100, using the EntityManager interface?

a.

```
EntityManager em = entityManagerFactory.createEntityManager();
TypedQuery<Object[]> query = em.createQuery(
    "SELECT p.name, s.name FROM Product p JOIN p.supplier s
WHERE p.price > 100",
    Object[].class
);
List<Object[]> result = query.getResultList();
```

b.

```
EntityManager em = entityManagerFactory.createEntityManager();
TypedQuery<Product> query = em.createQuery(
    "SELECT p FROM Product p WHERE p.price > 100",
    Product.class
);
List<Product> result = query.getResultList();
```

c.

```
EntityManager em = entityManagerFactory.createEntityManager();
TypedQuery<Object[]> query = em.createQuery(
    "SELECT p.name, p.supplier.name FROM Product p WHERE
p.price > 100",
    Object[].class
);
List<Object[]> result = query.getResultList();
```

d.

```
EntityManager em = entityManagerFactory.createEntityManager();
TypedQuery<Object[]> query = em.createQuery(
    "SELECT p.name, p.supplier FROM Product p WHERE p.price >
100",
    Object[].class
);
List<Object[]> result = query.getResultList();
```

Question 3

Which of the following statements correctly retrieves an entity of type Customer with a primary key of 101 using the EntityManager interface in JPA?

a. Customer customer = em.get(Customer.class, 101);

b. Customer customer = em.find(Customer.class, 101);

c. Customer customer = em.load(Customer.class, 101);

d. Customer customer = em.retrieve(Customer.class, 101);

Question 4

In the SQL:2011 standard, what does the keyword "FINAL" signify when defining a User-Defined Type (UDT)?

a. It allows the UDT to be used as a base type for other UDTs.

b. It specifies that the UDT cannot be subclassed.

c. It indicates that the UDT is only a placeholder and cannot be used to create actual instances.

d. It restricts the UDT from being used in views or tables.

Question 5

Given that the following SQL statements have been executed on a PostgreSQL database:

```
CREATE TYPE Product AS ( name text, category text, price
numeric );
CREATE TABLE Warehouse ( product Product, stock integer );

CREATE FUNCTION totalStockValue( Product, integer ) RETURNS
numeric AS
$$
    SELECT $1.price * $2 AS totalValue;
$$ LANGUAGE SQL;
```

Which one of the following SELECT statements will return a list of products along with their category, stock quantity, and the total stock value from the Warehouse table?

a.
```
SELECT product.name AS product_name, product.category AS
category,
       product.price AS unit_price, stock,
totalStockValue(product, stock)
FROM   Warehouse;
```

b.
```
SELECT (Warehouse.product).name AS product_name,
(Warehouse.product).category AS category,
       (Warehouse.product).price AS unit_price, stock,
totalStockValue(Warehouse.product, stock)
FROM   Warehouse;
```

c.
```
SELECT name AS product_name, category AS category,
       price AS unit_price, stock,
totalStockValue(product.name, stock)
FROM   Warehouse;
```

d.
```
SELECT product.name AS product_name, product.category AS
category,
       product.price AS unit_price, stock,
totalStockValue(Warehouse.product, stock)
FROM   Warehouse;
```

Question 6

Study the following code. What data type is `inventoryItem`?

```
CREATE TYPE inventoryItem AS (
name text,
supplierID integer,
price numeric);
```

a. Numeric

b. Enumerated

c. Text

d. Composite

Question 7

Which of the following statements about trigger functions in PostgreSQL is correct?

a. A trigger function can only be executed before an operation like INSERT, UPDATE, or DELETE occurs.

b. A trigger function can access both the old and new values of a row in the case of an INSERT operation.

c. A trigger function can only be attached to a table and cannot be attached to views.

d. A trigger function can execute after INSERT, UPDATE, or DELETE operations and can access the old row values for UPDATE and DELETE.

Question 8

Study the following XML file and answer questions 8 and 9 that follow:

```
<WAREHOUSE>
   <DEVICE type="laptop">
      <BRAND>Apple</BRAND>
      <MODEL>MacBook Pro</MODEL>
      <PRICE>2500</PRICE>
      <STOCK>50</STOCK>
   </DEVICE>
   <DEVICE type="smartphone">
      <BRAND>Samsung</BRAND>
      <MODEL>Galaxy S21</MODEL>
      <PRICE>799</PRICE>
      <STOCK>120</STOCK>
   </DEVICE>
   <DEVICE type="laptop">
      <BRAND>Dell</BRAND>
      <MODEL>XPS 13</MODEL>
      <PRICE>1800</PRICE>
      <STOCK>30</STOCK>
   </DEVICE>
   <DEVICE type="smartphone">
```

```
        <BRAND>Apple</BRAND>
        <MODEL>iPhone 12</MODEL>
        <PRICE>999</PRICE>
        <STOCK>200</STOCK>
    </DEVICE>
</WAREHOUSE>
```

Figure 1: devices.xml file

Which XQuery expression will return the brands of all laptops with a stock count less than 40?

a.

```
for $d in /WAREHOUSE/DEVICE[@type="laptop"]
where $d/STOCK < 40
return $d/BRAND
```

b.

```
for $d in /WAREHOUSE/DEVICE
where $d/@type = "laptop" and $d/STOCK < 40
return $d/BRAND
```

c.

```
for $d in /WAREHOUSE/DEVICE
where $d/STOCK < 40
return $d/BRAND[@type="laptop"]
```

d.

```
for $d in /WAREHOUSE/DEVICE[STOCK < 40]
where $d/@type = "laptop"
return $d/BRAND
```

Question 9

Which XQuery query will list the models and prices of all devices that are not smartphones?

a.

```
for $d in /WAREHOUSE/DEVICE
where $d/@type != "smartphone"
return $d/MODEL, $d/PRICE
```

b.
```
for $d in /WAREHOUSE/DEVICE
where $d/@type != "laptop"
return $d/MODEL, $d/PRICE
```

c.
```
for $d in /WAREHOUSE/DEVICE[@type!="smartphone"]
return $d/MODEL, $d/PRICE
```

d.
```
for $d in /WAREHOUSE/DEVICE[@type="laptop"]
return $d/MODEL, $d/PRICE
```

Question 10

The `table_to_xml` mapping function in PostgreSQL is used to:

a. Create an XML value containing an XML comment with the specified text.

b. Map the content of the named table.

c. Return the XML Schema documents describing the mappings.

d. Map the table columns to the corresponding XML schema.

Question 11

_____ refers to the ability of an operation to be applied to different types of objects and is a key feature of object-oriented programming.

Question 12

True/False: When using JPA, every operation on a database is associated with an EntityManager.

Question 13

Study the following code snippet below. Write the method call is needed to delete an entity from the database in the following code (do not any spaces in your answer).

```
Product product = em.find(Product.class, 1L);

__________
em.getTransaction().commit();
```

Question 14

Assume an Order entity with id = 100 and status = "Pending" exists in the ObjectDB database. The following code is executed:

```
Order order = em.find(Order.class, 100L);
em.getTransaction().begin();
order.setStatus("Completed");
em.getTransaction().commit();

Order o = em.find(Order.class, 100L);
System.out.println(o.getStatus());
```

What will be the output of the code?

Question 15

Study the following code:

```
CREATE OR REPLACE FUNCTION divide_numbers(a NUMERIC, b
NUMERIC) RETURNS NUMERIC AS $$
BEGIN
  IF b = 0 THEN
    ____________ 'Division by zero is not allowed!';
  END IF;
  RETURN a / b;
END;
$$ LANGUAGE plpgsql;
```

Complete the line of code such that an exception is properly caught

Question 16

Two types of SQL user defined functions are row functions and _____ functions.

Question 17

According to the SQL:2011 standard, a UDT may be a distinct type or a _____ type.

Question 18

The abbreviation XSD stands for_____

Question 19

Study the following code snippet:

```
<xs:emelement name = "phone_number" type = "xs:int" />
```

Name the XML schema data type that is shown in the above code. Enter the name of the type only.

Question 20

What is the name of the XML component that identifies node type, e.g. element name or function text()?

Section B Application and Theory - Based Questions (40 Marks)

(Spend approximately 30 minutes in this section)

Question 21 (15 Marks)

Object DBMS

(a) Define the following terms

(i) OO data model.

(ii) OID

(4 Marks)

(b) Explain the differences between SQL and JPQL. With the aid of an example, explain why JPQL is considered more "object-oriented" friendly than SQL. (4 Marks)

(c) You have the following JPQL query that retrieves all products with their associated categories:

```
Query q1 = em.createQuery("SELECT i FROM Invoice i JOIN
i.customer c WHERE c.country = :country");
```

(i) Convert the above JPQL query to a `TypedQuery` using the JPA API in Java. Assume that Invoice is a JPA entity class and you want to retrieve a list of `Invoice` entities. (3 Marks)

(ii) Explain two key benefits of using `TypedQuery` over using a basic `Query` in JPA. (4 Marks)

Question 22 (15 Marks)

ORDBMS

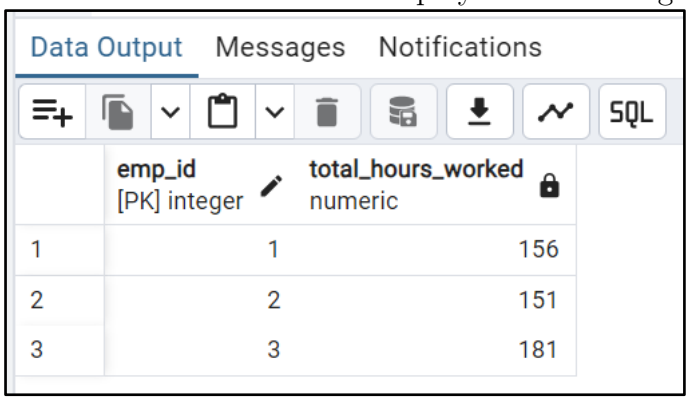
Study the following SQL statements and answer the following questions:

```
CREATE TABLE employee_hours (
    emp_id serial PRIMARY KEY,
    hours_worked_per_week numeric[]
);

INSERT INTO employee_hours (hours_worked_per_week) VALUES
    (ARRAY[40, 42, 38, 36]),
    (ARRAY[35, 37, 39, 40]),
    (ARRAY[45, 43, 47, 46]);
```

You have executed the above statements in PostgreSQL, and they ran successfully.

(a) Complete the following function such that it calculates the total number of hours worked over all weeks and displays the following result:



	emp_id [PK] integer	total_hours_worked numeric
1	1	156
2	2	151
3	3	181

Figure 2 - Query output

```
CREATE FUNCTION total_hours_worked(numbers numeric[])
RETURNS ____________________
DECLARE
    total numeric := 0;
    num numeric;
BEGIN
    _____________ num IN ARRAY numbers
    LOOP
        total := total + num;
    _____________;

    _________________;
END;
$$ ___________________;
```

(10 marks)

(b) Write the Select statement query that will output the result shown in Figure 2. (2 Marks)

(c) Provide an example where a PL/pgSQL function would be more appropriate than a SQL function and explain why. (3 Marks)

Question 23 (10 Marks)

Semi structured and XML

You are working with an XML dataset that contains information about an airline's flight schedule. The XML includes details such as flight number, departure city, destination city, departure time, arrival time, and whether the flight is on time or delayed. Below is a sample of the XML structure:

```
<FLIGHTSCHEDULE>
   <FLIGHT>
      <FLIGHTNO>AA101</FLIGHTNO>
      <DEPARTURECITY>New York</DEPARTURECITY>
      <DESTINATIONCITY>London</DESTINATIONCITY>
      <DEPARTURETIME>2024-09-15T08:30:00</DEPARTURETIME>
      <ARRIVALTIME>2024-09-15T20:30:00</ARRIVALTIME>
      <STATUS>On Time</STATUS>
   </FLIGHT>
   <FLIGHT>
      <FLIGHTNO>BA202</FLIGHTNO>
      <DEPARTURECITY>London</DEPARTURECITY>
      <DESTINATIONCITY>Paris</DESTINATIONCITY>
      <DEPARTURETIME>2024-09-15T10:00:00</DEPARTURETIME>
      <ARRIVALTIME>2024-09-15T12:00:00</ARRIVALTIME>
      <STATUS>Delayed</STATUS>
   </FLIGHT>
   <FLIGHT>
      <FLIGHTNO>DL303</FLIGHTNO>
      <DEPARTURECITY>Los Angeles</DEPARTURECITY>
      <DESTINATIONCITY>Tokyo</DESTINATIONCITY>
      <DEPARTURETIME>2024-09-15T16:45:00</DEPARTURETIME>
      <ARRIVALTIME>2024-09-16T08:00:00</ARRIVALTIME>
      <STATUS>On Time</STATUS>
   </FLIGHT>
</FLIGHTSCHEDULE>
```

Using FLWOR queries in XQuery, answer the following questions:

(a) Write a XQuery FLWOR expression to return a list of all flights that are currently delayed. Display the flight number, departure city, and destination city. (3 Marks)

(b) Write a XQuery FLWOR expression to return all flights sorted by their departure time in ascending order. Display the flight number, departure time, and destination city. (5 Marks)

(c) Explain the purpose of the "Where" clause in a FLWOR query and how it filters results. (2 Marks)

Section C: Scenario Based Questions (40 Marks)

(Spend approximately 45 minutes in this section)

Question 24 (15 Marks)

OODBMS

Study the following scenario and answer the following questions:

You are tasked with developing a pet adoption management system for a local animal shelter. Each pet in the system is represented as an entity with the following attributes:

name: A string representing the name of the pet.
species: A string representing the species of the pet (e.g., dog, cat, rabbit).
age: An integer representing the age of the pet.
adopted: A boolean value indicating whether the pet has been adopted or not.
The system is built using JPA (Java Persistence API) and ObjectDB as the database. The goal is to implement some CRUD operations to manage the pets in the shelter.

Below is an example of the main method used to test the functionality:

```
public class Main {
    public static void main(String[] args) {
        PetDAO petDAO = new PetDAO();
        Pet newPet = new Pet("Buddy", "dog", 3, false);
        petDAO.addPet(newPet);
        // Rest of the program
    }
}
```

Source: Makura S.M (2024)

(a) Write the Pet entity class that includes the required members, using JPA annotations to map the class to the ObjectDB table. (PS: there is no need to include the getters and setters) (5 Marks)

(b) Implement the `addPet` method in the `PetDAO` class, which takes a `Pet` object as input and stores it in the database. The method should ensure that the `pet` is persisted correctly. Provide the method implementation. (3 Marks)

You would like to view a list of all pets currently available for adoption. To do this,

you need to write a method `listAvailablePets` that retrieves all pets where the adopted field is false.

(c) Explain why using a JPQL (Java Persistence Query Language) query is necessary for this operation and provide the query you would use. (3 Marks)

(d) Write the implementation of the `listAvailablePets` method, which returns a list of pets available for adoption. (4 Marks)

Question 25 (15 Marks)

ORDBMS

Study the following scenario and answer the following questions:

> Palesa has asked for your help to automate the process of updating the stock in the inventory table whenever a new order is placed. Specifically, when an order is inserted, the stock of the corresponding item in the inventory table should be reduced by the quantity ordered. If the stock falls below 5, the system should set the `order_status` of the order to `LowStockOrder;` otherwise, it should be set to `InStockOrder.);`
>
> Source: Makura S.M (2024)

Questions:

(a) Complete the trigger function below that updates the stock in the inventory table and sets the appropriate `order_status` in the orders table based on the stock level of the ordered item.

(7 Marks)

```
CREATE FUNCTION update_stock() RETURNS ______ AS $$
DECLARE
    current_stock INT;
BEGIN
    SELECT stock INTO ___________
    FROM __________
    WHERE item_id = NEW.item_id;

    IF current_stock >= NEW.quantity THEN
        UPDATE inventory
        SET stock = stock - NEW.quantity
        WHERE __________ = NEW.item_id;
```

```
        IF current_stock - NEW.quantity < 5 THEN
            _____________ = 'LowStockOrder';
        ELSE
            _____________ = 'InStockOrder';
        END IF;
    ELSE
        _____________ 'Insufficient stock for item_id %',
NEW.item_id;
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

(b) Explain the advantages and disadvantages of using triggers in object relational database systems. (4 Marks)

(c) Complete the following trigger such that it calls the trigger function `update_stock` before a new order is inserted. (4 Marks)

```
CREATE __________ update_stock_trigger
___________ ON orders
FOR ____________
EXECUTE __________ update_stock();
```

Question 26 (10 Marks)

Semi-structured Data and XML

Study the following scenario and answer the following questions:

You are working with an employee performance database where the performance evaluations are stored as XML data, along with an array of quarterly performance scores in a PostgreSQL database. The company tracks various aspects of an employee's work, such as completed projects, feedback ratings, and salary, using a mixture of XML and array data types.

The following SQL statements were successfully executed in PostgreSQL:

```
CREATE TABLE EmployeePerformance (
    empId CHAR(4) PRIMARY KEY,
    empName VARCHAR(100),
    performanceData XML,
    quarterlyScores numeric[]
```

```
);

INSERT INTO EmployeePerformance (empId, empName,
performanceData, quarterlyScores)
VALUES ('E001', 'John Doe',
    XML('<PERFORMANCE year="2023">
            <PROJECTS>5</PROJECTS>
            <RATING>4.5</RATING>
            <SALARY>60000</SALARY>
        </PERFORMANCE>'),
    ARRAY[8.5, 7.0, 9.0, 8.0]
);

INSERT INTO EmployeePerformance (empId, empName,
performanceData, quarterlyScores)
VALUES ('E002', 'Jane Smith',
    XML('<PERFORMANCE year="2023">
            <PROJECTS>6</PROJECTS>
            <RATING>4.8</RATING>
            <SALARY>65000</SALARY>
        </PERFORMANCE>'),
    ARRAY[9.0, 8.5, 8.0, 9.5]);
```

The table stores:

empId: A unique identifier for each employee.
empName: The name of the employee.
performanceData: XML data that stores the number of completed projects, the performance rating, and the salary for the year.
quarterlyScores: An array that contains the employee's performance scores for each quarter.

Source: Makura S.M (2024)

(a) Write a query to extract the employee's performance rating from the performanceData XML field for each employee. Use the xpath() function to do this. (3 Marks)

(b) Complete the following PL/pgSQL function that calculates the average performance score for each employee based on the quarterlyScores array. The function should return the average score. (3 Marks)

```
CREATE FUNCTION calculate_avg_score(scores numeric[])
RETURNS numeric AS $$
```

```
DECLARE
    total numeric := 0;
    count int := 0;
    average numeric;
BEGIN
    FOR score IN SELECT unnest(_____)
    LOOP
        total := total + score;
        count := count + 1;
    END LOOP;

    IF count > 0 THEN
        average := _______;
    ELSE
        average := NULL;
    END IF;
    RETURN _______;
END;
$$ LANGUAGE plpgsql;
```

(c) Write a query using the xpath() function to extract the salary from the performanceData XML field for employees who have a performance rating greater than 4.6. The query should return the employee ID, name, and salary. (4 Marks)

End of Semester Test Paper